

Jiahui Huang

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Education

PhD in Pure Mathematics University of Waterloo, Canada Supervisor: Matthew Satriano	Sept 2023 – present
Master in Mathematics ETH Zürich, Switzerland Supervisors: Arkadij Bojko, Rahul Pandharipande	Sept 2021 – April 2023
Bachelor in Pure Mathematics University of Waterloo, Canada	Sept 2017 – June 2021

Publications and Preprints

<i>The dimension threshold for Batyrev's non-negativity conjecture on stringy Hodge numbers.</i> Joint with Matthew Satriano. <i>Preprint.</i> arXiv:2608.07836	2026
<i>A cohomological interpretation for stringy Hodge numbers.</i> Joint with Matthew Satriano and Jeremy Usatine. <i>Preprint.</i> arXiv:2602.19344	2026
<i>Contact loci of semihomogeneous singularities.</i> Joint with Eduardo de Lorenzo Poza. <i>Preprint.</i> arXiv:2506.18805	2025
<i>Equivariant Segre and Verlinde invariants for Quot schemes.</i> Joint with Arkadij Bojko. <i>Moduli.</i>	2025
<i>What fraction of an S_n-orbit can lie on a hyperplane?</i> Joint with David McKinnon and Matthew Satriano. <i>Linear Algebra and Its Applications.</i>	2021

Selected Awards

PMATH Outstanding Teaching Assistant Award	2025
Women in Math Mentorship Award	2024 – 2026
Undergraduate Research Assistanceship	2019 – 2020
President's Scholarship	2018
J.D. Kalbfleisch Scholarship for International Students	2017 – 2021

Teaching

MATH 127 Calculus for Sciences	2025
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Talks

<i>Contact Loci of Semihomogeneous Singularities.</i> Florida State University.	2025
<i>Arc-Floer conjecture for homogeneous singularities.</i> Ontario Graduate Mathematics Conference.	2024
<i>Arc-Floer conjecture for double point singularities.</i> CMS Summer Meeting.	2024
<i>Equivariant Segre and Verlinde invariants for Quot schemes.</i> ETH Zürich.	2023

Conference Activities

The 5th National Algebraic Geometry Conference, University of Peking.	2026
Chicago-Tokyo Algebraic Geometry Workshop, University of Illinois Chicago.	2026
Singularities in Algebraic Geometry Spring School, University of Rennes.	2025
Ontario Graduate Mathematics Conference, University of Waterloo.	2024
Canadian Mathematical Society Summer Meeting, University of Saskatchewan.	2024
Equivariant Methods in Geometry, Cambridge University.	2024
Algebraic Geometry Summer School, Fudan University.	2023
Algebraic Geometry Summer School, Fudan University.	2021